

# Nazmul Haque Turja

nhturja@udel.edu · Newark, Delaware, USA

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## Education

### PhD in Electrical and Computer Engineering

University of Delaware

Newark, Delaware · August 2026 – Present

Research focus: Hardware security.

### MS in Electrical Engineering

New Mexico State University

Las Cruces, New Mexico · August 2021 – May 2023

Master's project: Utilizing CPI Breakdown for Bottleneck Analysis of Power9 Processor Performance

### BS in Electrical and Electronic Engineering

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh · July 2014 – April 2019

Major: Communication Engineering. Thesis: A Secured Offline Online Approach for Internet of Things (IoT) Using Real-time Database.

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## Experience

### Graduate Research Assistant

University of Delaware · Department of ECE

Newark, Delaware · Upcoming · starts October 1, 2026

### GPU Logic Design Engineer

Intel Corporation

Folsom, California · July 2023 – September 2026

- Conducted functional pre-silicon verification for discrete, integrated, and HPC GPUs, focusing on 3D render and compute pipelines.
- Owned end-to-end validation of Intel Xe-Core ray tracing and texture sampling features, including test planning, test development, and coverage closure.
- Used UVM methodologies and debugged regression failures in simulation and emulation, collaborating with design and architecture teams.

### Graduate Assistant

New Mexico State University · Department of ET and SE

Las Cruces, New Mexico · January 2023 – May 2023

- Instructed Communication Systems I (ET 314) theory and lab sessions and mentored students on assignments and projects.

## Graduate Technical Intern

Intel Corporation

Folsom, California · May 2022 – December 2022

- Addressed power flow analysis (ptpx) issues for low-power Intel Xe-Core Sampler designs.
- Managed regressions for the Meteor Lake Sampler CLT out-of-order feature and identified two silicon hangs.
- Developed UVM testbenches and worked with gnr, elab, and runsim tools.

## Graduate Research and Teaching Assistant

New Mexico State University · Department of ECE

Las Cruces, New Mexico · August 2021 – May 2022

- Applied machine learning to basic block count prediction with NVBit-supported PPT-GPU for PolyBench, Rodinia, Pannotia, and LULESH workloads.
- Collaborated with Los Alamos National Laboratory on reuse distance and memory tracing toolkits for PPT-GPU, and with Sandia National Laboratories on LDMS sampling for IBM Power9.
- Taught electronics, digital circuit design, and VHDL to more than 50 students.

## Adjunct Faculty

BRAC University · Department of CSE

Dhaka, Bangladesh · February 2021 – September 2021

- Taught VLSI Design (CSE 460) and Digital Electronics and Pulse Techniques (CSE 350), including labs with Proteus, Quartus II, Microwind, and ModelSim.
- Worked on IoT applications for healthcare and agriculture under Dr. Farhad Hossain at BUET.

## Semiconductor Engineer

Nelsite Inc. Ltd.

Fukuoka, Japan · November 2019 – April 2020

- Developed embedded systems with 32-bit ARM Cortex-M4 microcontrollers, Keil, and embedded C.
- Received on-the-job training in semiconductor fabrication and material characterization.

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## Technical skills

Languages: CUDA, OpenCL, Python, Embedded C/C++, VHDL, SystemVerilog, Go, Lex, Yacc, LaTeX.

Software and tools: Verdi, UVM, NumPy, Pandas, TensorFlow, Keras, scikit-learn, Linux, Eagle, Proteus, Quartus II, Cadence, OpenCV.

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## Selected coursework

Computer engineering: Advanced Computer Architecture, Application of Parallel Programming, ASIC Design, Hardware and Software Co-design, Compiler Transition, Computer Architecture, VLSI, Microprocessor and Interfacing, Analog and Digital Electronics.

Artificial intelligence: Application of Machine Learning, Deep Learning, Random Signal Analysis, Linear Algebra, Probability and Statistics.

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## Awards

Battle of Hardware (IoT) — Champion, CSE Festival 2018, BUET.

## Publications

Nazmul Haque Turja. Utilizing CPI Breakdown for Bottleneck Analysis of Power9 Processor Performance. Master's project report, New Mexico State University, 2023.

Hamdy Abdelkhalik, Shamminuj Aktar, Yehia Arafa, Atanu Barai, Gopinath Chennupati, Nandakishore Santhi, Nishant Panda, Nirmal Prajapati, Nazmul Haque Turja, Stephan Eidenbenz, Abdel-Hameed Badawy. BB-ML: Basic Block Performance Prediction using Machine Learning Techniques. 29th IEEE International Conference on Parallel and Distributed Systems (ICPADS), 2023.

Nahian Ibn Hasan, Md. Tasnimul Hasan, Nazmul Haque Turja, Rishad Raiyan, Shuvagata Saha, Md Farhad Hossain. A Cyber-Secured MQTT based Offline Automation System. International Conference on Wireless Communications Signal Processing and Networking (WiSPNET), 2019.

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## Selected projects

### RTL Medic

Research on hardware design and verification using LLMs and AI agents. Ongoing collaborative research.

### Hardware validation with ASMD

State-machine reasoning for hardware verification. Completed a hardware design-validation project.

### Cache memory simulator

A C++ simulator for exploring cache organizations. Built a configurable cache simulation implementation.

### MIPS compiler project

Exploring compilation with abstract syntax trees. Completed the course implementation.

### An 8-bit computer

A Simple As Possible computer simulated in Proteus. Designed and simulated the computer.

### General-purpose shift register

A small digital design with multiple data-movement modes. Produced the register design with the three operating modes.

### Indoor wayfinding wearable

A wearable project for people with Alzheimer's disease. Built a wearable prototype.